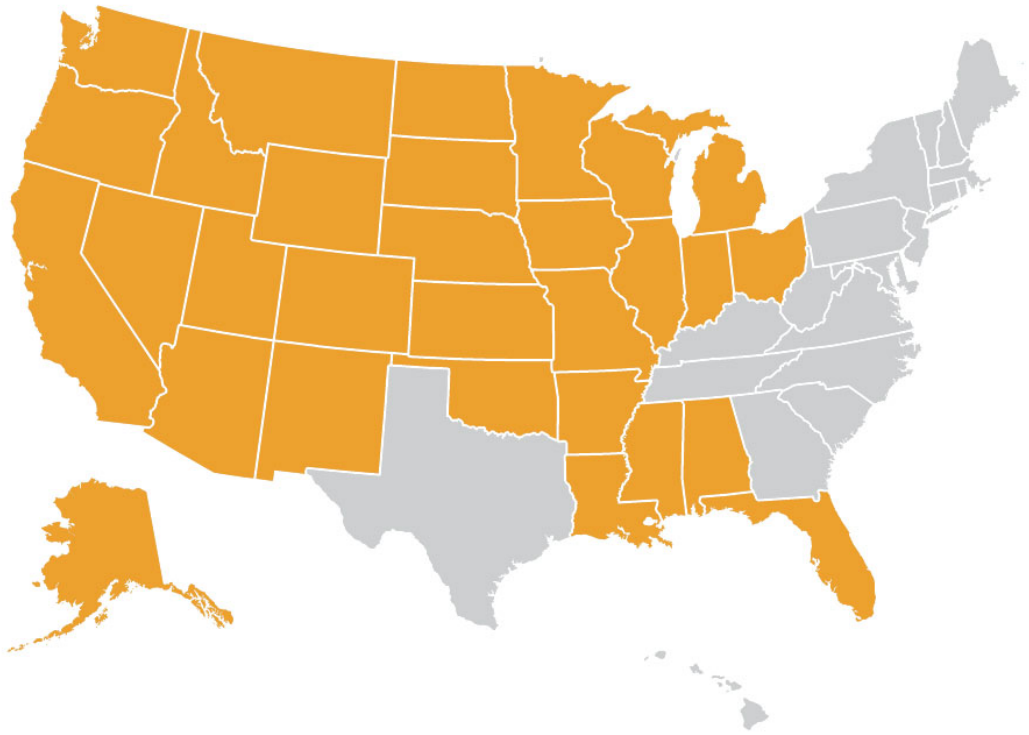
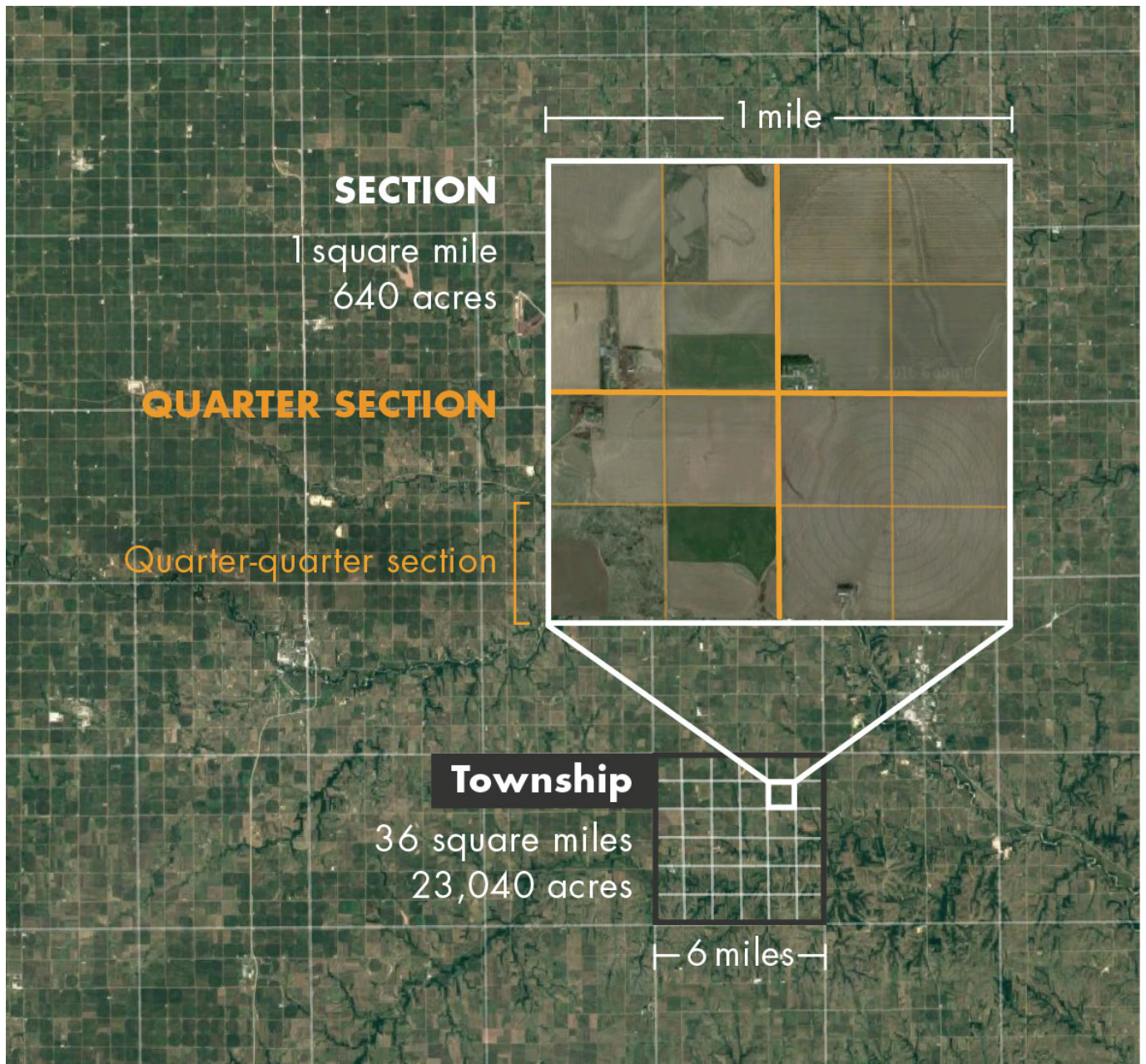


The Rectangular Government Survey System

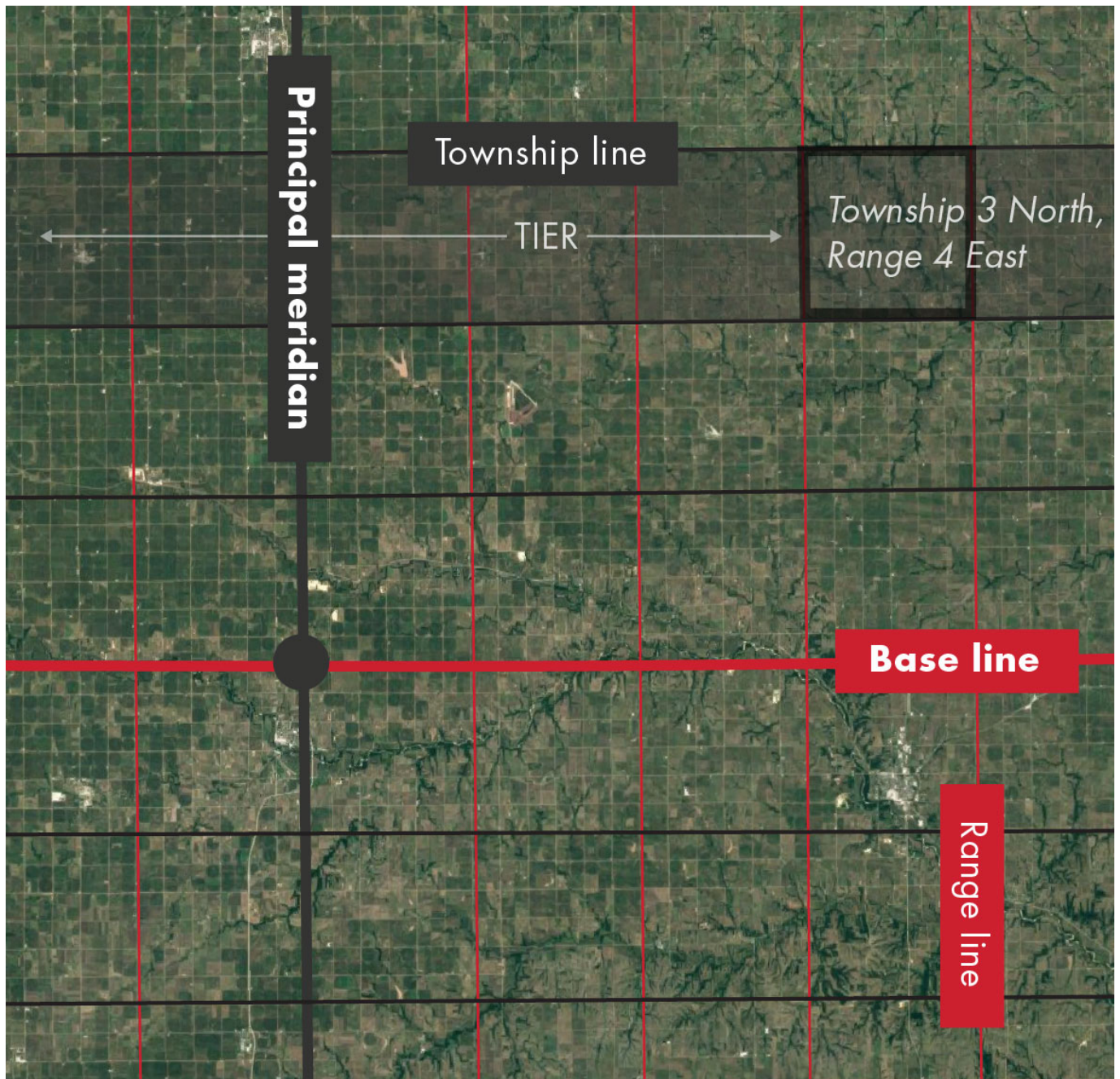
The rectangular government survey system (also known as the Public Land Survey System (PLSS), because it was originally used to survey federally owned blocks of land) is regulated by the U.S. Department of the Interior, Bureau of Land Management.



The rectangular government survey system is only used in 30 western and southern states; it is not used by east coast states except Florida; it is also not used by Texas. Some states, like Missouri, used this method of land surveying in their original land surveys, as opposed to the metes and bounds method.



The PLSS system is actually a series of surveys that divide land into townships that are six miles by six miles square (for a total area of 36 square miles or 23,040 acres). Townships are further subdivided into 36 one-mile-square (640 acres) sections (each section is one square mile or 640 acres) then into **quarter sections**, **quarter-quarter sections**, or irregular government lots.

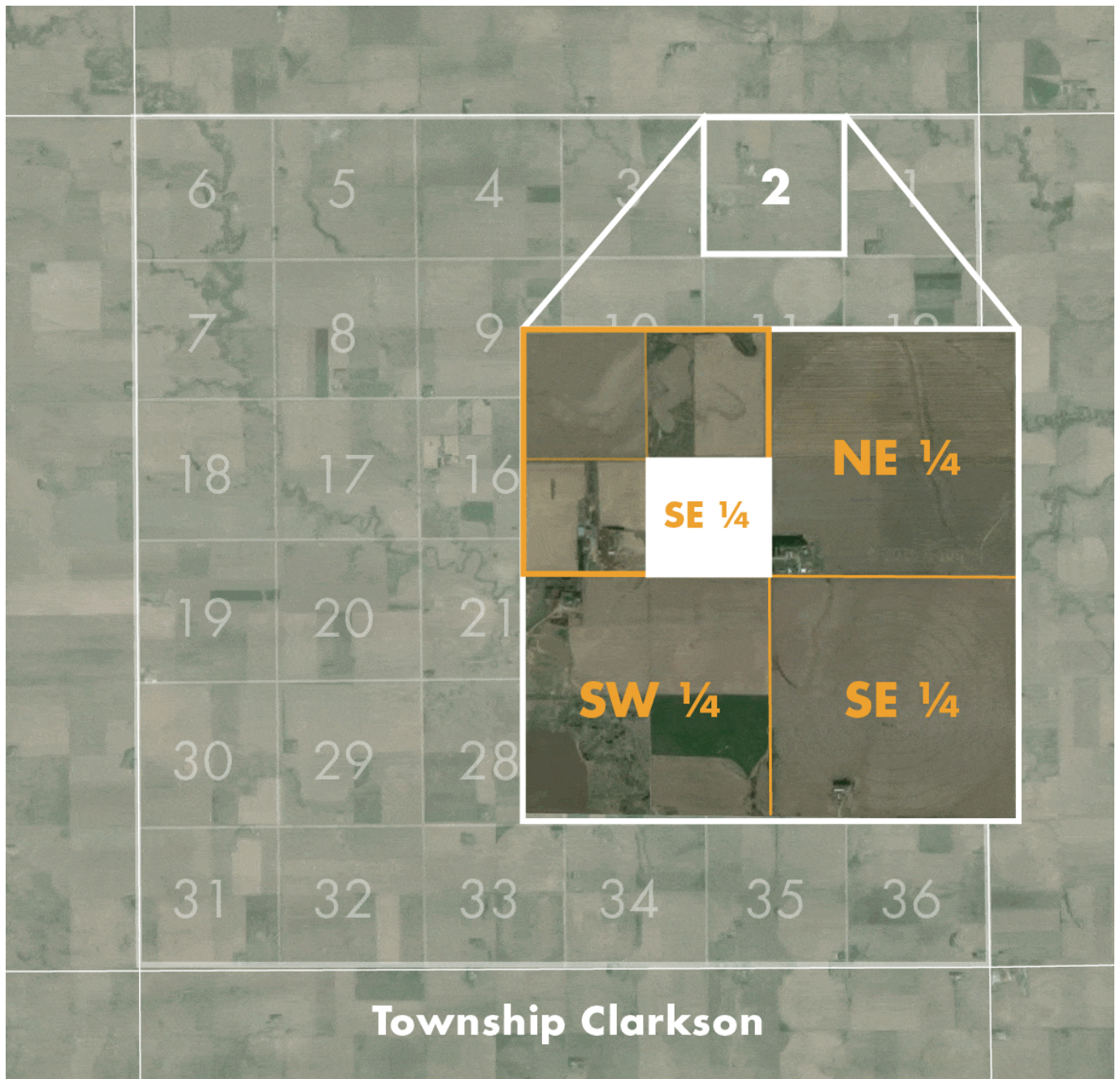


Beginning at an initial point, townships are surveyed in all directions. The north-south line that runs through the initial point is a true meridian called the **principal meridian**. There are 37 principal meridians in the United States, each referred to by name or number. For example, all land in Missouri is referenced to the Fifth Principal Meridian. The east-west lines that cross the principal meridians are called **base lines**; the north-south lines that run parallel to the principal meridians in six-mile-wide strips are called **range lines**. Lines that run east and west, parallel to the base line and six miles apart, are called **township lines**. The township lines form strips of land known as **tiers**.

***Here's an example:** A township might be identified as Township 3 North, Range 4 East, which would mean that it was the 3rd tier of townships north of a base line and in the fourth column of townships east of a principal meridian.*



Here's a simple layout showing how a township is designated. Note that the numbers do not read left to right, but start at the northeastern corner and follow a zigzag pattern. You may see an exam question or two about this, so be sure you understand how the grid is laid out.

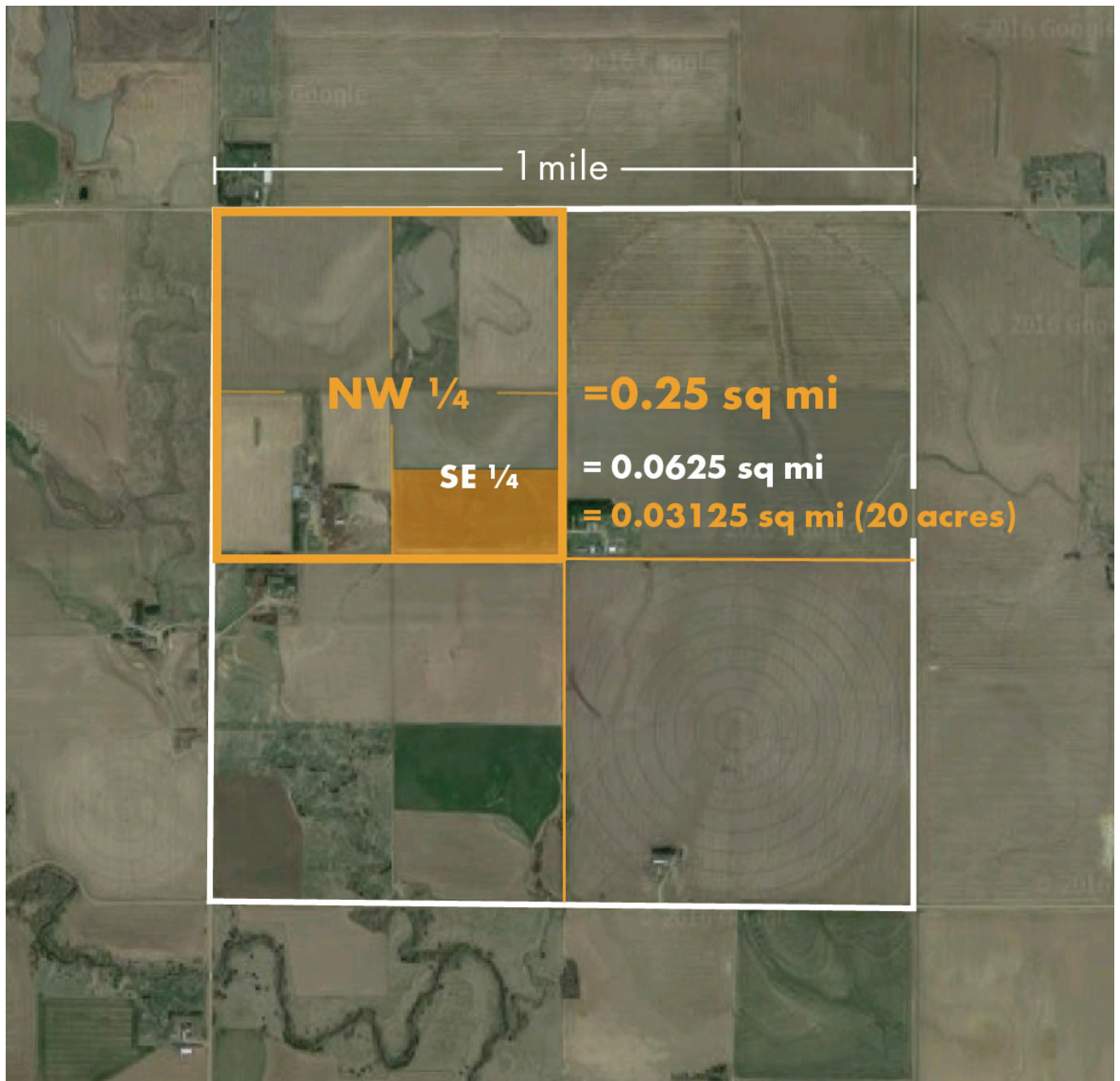


Look at the highlighted parcel of land in the fictional township of Clarkson. This would be described as **S½ SE¼ NW¼ Section 2, Township Clarkson.**

If you were saying it out loud, you'd say: "The southern half of the southeast quarter of the northwest quarter of Section 2 of Township Clarkson."

When land is mapped on the rectangular survey system, you read the directions beginning from right to left. Therefore, in Section 2, we look for the northwest 14, and from that we find the southeast 14, and from there we find the south 12, which is the highlighted portion.

Note: Legal descriptions using the PLSS system include the State, Principal Meridian name, Township and Range designations with directions and the section number.



How large is this parcel?

- If we know that each side of a section is 1 mile long ($1 \times 1 = 1$ square mile), then the NW 14 is 0.5×0.5 miles, or 0.25 square miles.
- One-fourth of that (the SE 14 of the NW 14) would be 0.0625 square miles.
- One-half of 0.0625 square miles (the S 12 of the SE 14) would be 0.03125 square miles, or 20 acres.
 One acre is 0.0015625 of a square mile, so: **$0.03125 \text{ sq mi} \div 0.0015625 = 20 \text{ acres}$**



How large is this parcel?

An easier way to compute this is to start with acres in the first place. As you know, the entire section is 640 acres. The shaded parcel, we can see, is $\frac{1}{32}$ of 640 acres. But if you didn't have the diagram to look at, you can easily see what portion of 640 acres we're talking about using the directions themselves:

$$12 \times 14 \times 14 = \frac{1}{32}$$

Now, you simply take 640 acres and divide it by 32 and you get 20 acres.

$$640 \text{ acres} \div 32 = 20 \text{ acres}$$